Comptia N+ Assignment

[N+ Network Fundamental and building networks]

Network Topologies

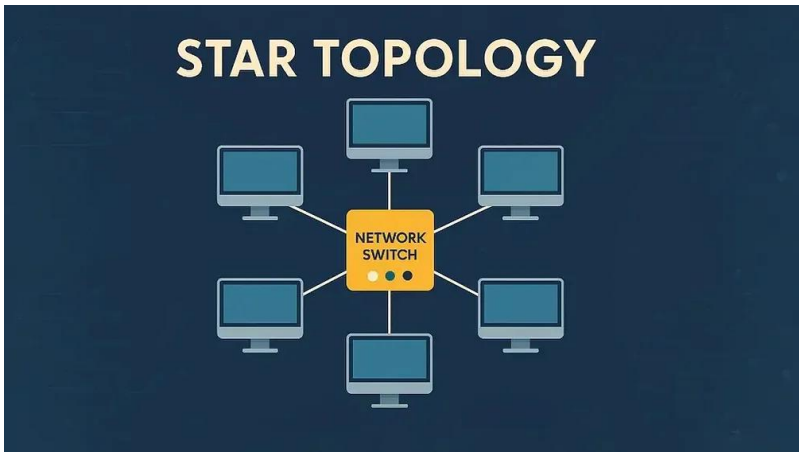
Task:

- **1. Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.**

  - **Aim**
    - To draw and understand Star, Bus, and Mesh network topologies and explain their uses with real-world examples.

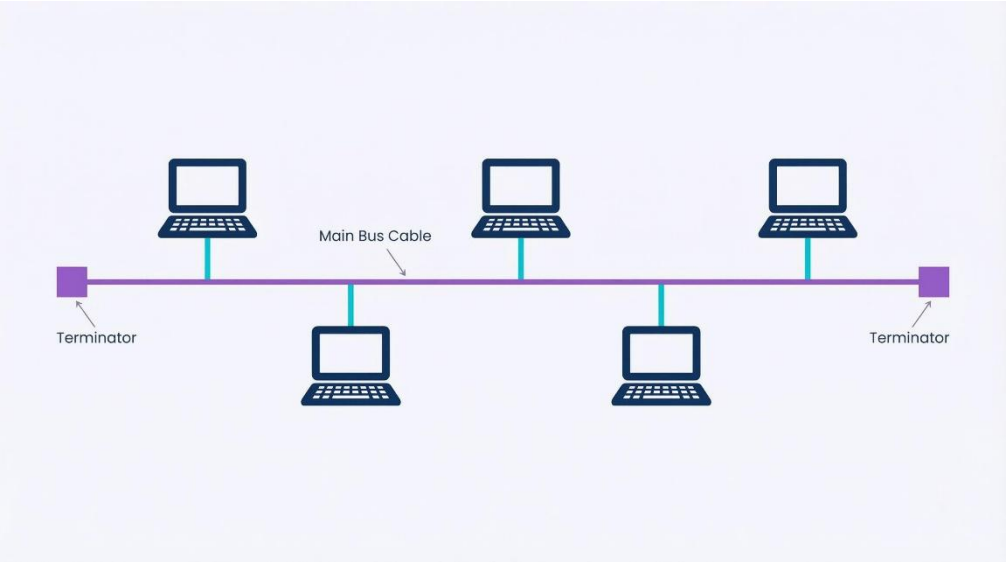
  - **Introduction**
    - A network topology is the physical or logical arrangement of devices and connections in a computer network. It describes how computers and other devices are connected and how they communicate with each other.
    - The three topologies discussed in this assignment are Star, Bus, and Mesh.

- **Star Topology**
- **Definition:** In a Star Topology, all devices are connected to a central device, usually a switch. All communication between connected devices passes through this central device.



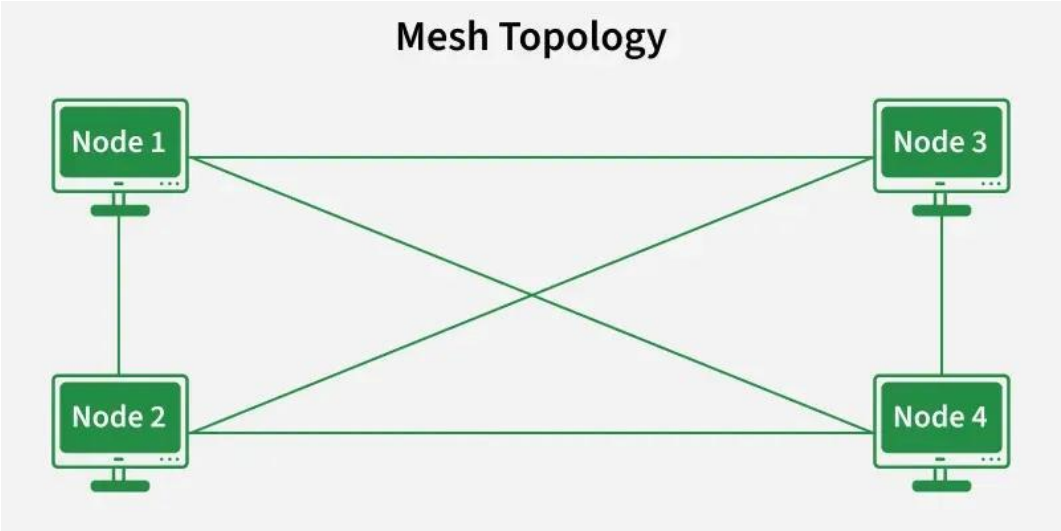
- **Real-world example: WhatsApp**
- A star-shaped local network could connect users' computers or devices to a central switch in an office where WhatsApp is used. The switch helps manage communication within the local network, while WhatsApp messages travel through the Internet to its servers.

- **Bus Topology**
- **Definition:** In a Bus Topology, all devices share a single main communication cable called the backbone. Terminators are placed at both ends of the cable to reduce signal reflections.



- **Real-world example: IPL Streaming**
- A shared bus network could distribute the same video stream to multiple devices in a small, controlled local setup. However, modern IPL streaming services generally use Internet infrastructure rather than a traditional bus topology.

- **Mesh Topology**
- **Definition:** In a Mesh Topology, devices are connected through multiple links. In a full mesh, every device has a direct link to every other device, providing alternative communication paths.



- **Real-world example: Flipkart**
- A mesh-like network could connect servers or network devices in an e-commerce infrastructure. Multiple paths can help maintain connectivity if one link fails, although Flipkart's actual internal network design is not assumed to be a full mesh.

- **Comparison of Network Topologies**

Feature	**Star**	**Bus**	**Mesh**
Structure	Central switch	Single backbone cable	Multiple interconnected links
Cost	Moderate	Usually low for small networks	Usually high

Reliability	Central device is a potential single point of failure	Backbone failure can affect the whole network	Multiple paths improve reliability
Common use	Office and home LANs	Small or legacy networks	Network backbones and resilient networks

❖ **2. Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).**

➢ **Aim**

- To compare Star, Bus, Ring, and Mesh network topologies based on their advantages, disadvantages, best use cases, and real-world application examples.

➢ **Introduction**

- A network topology describes how computers and other devices are connected in a network. Different topologies have different costs, levels of reliability, and maintenance requirements. The choice of topology depends on the size and purpose of the network.

➢ **Comparison Table**

Topology	Advantages	Disadvantages	Best Use Case	App Example

Star	1. Easy to install. 2. Easy to troubleshoot. 3. Failure of one device's cable usually affects only that device. 4. Easy to add new devices.	1. Requires more cable than Bus. 2. Needs a central switch or hub. 3. Central device failure can stop the network. 4. Installation cost can be higher than Bus.	Home, college and office LANs.	Instagram: Users' devices can connect through a star-shaped office or home LAN to access Instagram services over the Internet.
Bus	1. Simple design. 2. Uses less cable in small networks. 3. Low installation cost for small setups. 4. Does not require a central switch.	1. Backbone cable failure can stop the network. 2. Performance decreases as traffic increases. 3. Troubleshooting cable faults can be difficult.	Small or legacy networks with few devices.	Zomato: A traditional bus network could connect devices in a small test lab used to develop or test a food-ordering application.

		4. Limited scalability and cable length.		
Ring	1. Each device has a defined connection to its neighbors. 2. Can provide orderly data transmission. 3. Can reduce collisions in token-based ring networks. 4. Performance can be predictable under controlled traffic.	1. A single link or device failure can disrupt a basic ring. 2. Adding or removing devices can interrupt communication. 3. Troubleshooting can be difficult. 4. Expansion and maintenance may be complex.	Specialized or legacy networks that need orderly access to the transmission medium.	Paytm: A ring topology could be used in a hypothetical test network for payment-system devices; this does not imply that Paytm uses a ring topology in its actual infrastructure.
Mesh	1. Provides multiple communication paths.	1. Expensive to install.	Critical networks, resilient backbones and	Instagram: A mesh-like network could connect

	2. Highly reliable when alternative paths are available. 3. Can continue operating when a link fails. 4. Supports redundancy and fault tolerance.	2. Requires many cables or wireless links. 3. Configuration and maintenance can be complex. 4. A full mesh becomes difficult to scale as devices increase.	systems that need high availability.	network devices or servers to provide alternative paths. Instagram's actual internal topology is not being claimed here.

❖ **3. Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.**

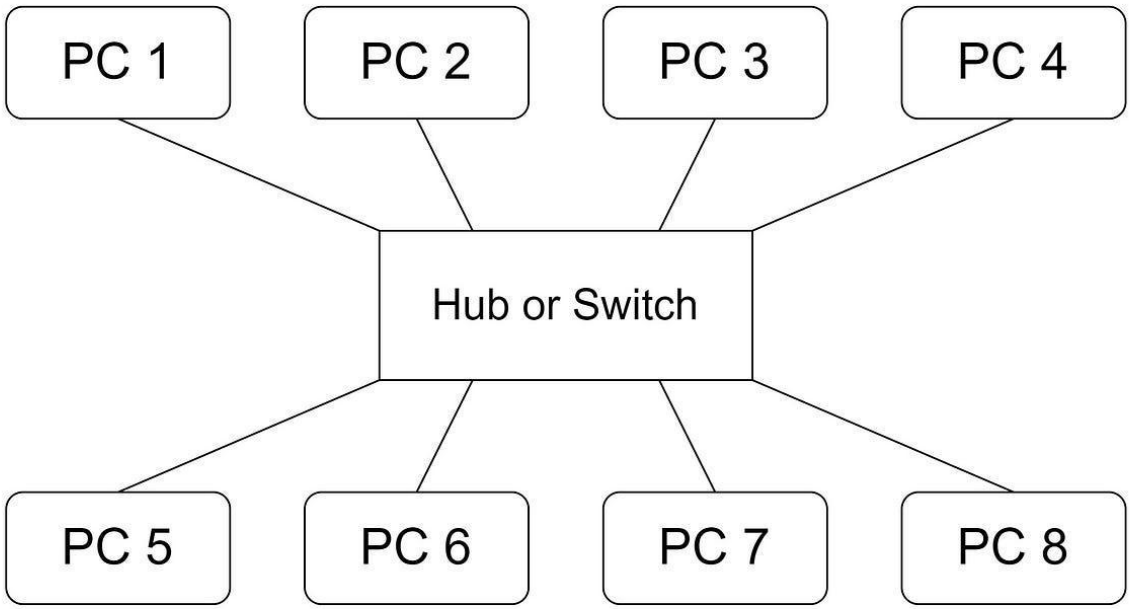
➢ **Chosen Topology: Mesh Topology**

➢ I would choose a **Mesh Topology** for the network infrastructure connecting multiplayer game servers because it provides multiple communication paths between network devices. These alternative paths can improve reliability and help maintain connectivity if one link fails. With suitable high-speed connections, it can support fast communication and reduce

delays between servers. However, a full mesh can be expensive and complex, so a partial mesh would be a more cost-effective choice for a large multiplayer game.

❖ **4. Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines. Consider factors like cost, ease of troubleshooting, and future expansion.**

➢ **Recommended Topology: Star Topology**



- **Explanation**
  - Star Topology is suitable for the warehouse because all eight computers connect to a central switch, making the network easy to manage and troubleshoot. It is cost-effective for a small network, and the failure of one computer or its cable usually does not affect the other computers. New computers can be added easily by connecting them to available switch ports.